



BT5153 Applied Machine Learning for Business Analytics

Group Project Proposal (Group 14)

Full Name	Matriculation Number
Phang Beng Keong	A0028103X
Phua Zhuo Wei	A0234313U
Sakthi Ram	A0156203R
Liu Yuandong	A0297814N

BUSINESS SCHOOL

DEPARTMENT OF INFORMATION SYSTEMS AND ANALYTICS

NATIONAL UNIVERSITY OF SINGAPORE

Motivation

1. Recruitment teams increasingly face a high-volume screening problem. For many roles, hundreds or even thousands of resumes are submitted, yet many older Applicant Tracking Systems (ATS) still rely heavily on keyword matching. This lexical approach fails to capture semantic meaning and transferable skills. For example, not being able to distinguish between "Java" the coffee and "Java" the language, or not recognizing that "ML" is synonymous with "Machine Learning". As a result, qualified candidates may be overlooked due to superficial mismatches, while recruiters expend significant time manually screening applications.
2. This project aims to design and evaluate a machine learning-based resume-job matching system that ranks candidates based on semantic suitability rather than exact keyword overlap. From a business analytics perspective, improving ranking quality can reduce screening effort and increase shortlist precision. More broadly, we seek to evaluate whether modern NLP approaches meaningfully outperform classical baselines in this applied recruitment context.

Problem Definition and Success Criteria

3. The task is a ranking problem: given a job description and a pool of resumes, the system ranks candidates by semantic suitability. Success is defined across both business and technical dimensions.
4. From a business perspective, improved ranking quality should increase shortlist precision; i.e., a higher proportion of truly suitable candidates appearing in the top positions, thereby reducing manual screening effort.
5. Technically, we evaluate performance using ranking metrics, specifically **Precision@10** and **Mean Reciprocal Rank (MRR)**. These are appropriate because recruiters typically review only the top results. Precision@10 measures the proportion of relevant candidates within the top 10, directly reflecting shortlist quality, while MRR captures how early the first relevant candidate appears, reflecting screening efficiency.
6. We target **Precision@10 ≥ 0.80** and **MRR ≥ 0.70** on a held-out test set to ensure the system consistently surfaces strong candidates near the top of the ranking.

Data and Methodology

7. To evaluate resume-job matching, we will compile approximately 2,000 resumes and job descriptions from publicly available sources (e.g., Kaggle and public job postings), with all personally identifiable information removed. The data will be split into training (70%), validation (15%), and test (15%) sets for consistent evaluation.

8. We will compare three approaches: (1) a lexical TF-IDF + cosine similarity baseline, (2) a semantic model using pre-trained Sentence-BERT embeddings, and (3) an LLM-assisted zero- or few-shot ranking approach. All models will be evaluated on the same held-out test set. We will select the simplest model that meets performance targets; more complex methods will be adopted only if they deliver statistically robust and practically meaningful improvements over the baseline within feasible computational constraints.

Design Boundaries and Guardrails

9. To ensure the evaluation reflects realistic deployment conditions, we impose the following constraints:

a. **Performance must be materially meaningful.**

Model comparison will not rely solely on incremental gains in Precision@10 or MRR. More complex models will be adopted only if they deliver substantively better ranking quality relative to simpler baselines, without disproportionate increases in computational cost or operational complexity.

b. **Interpretability as a requirement.**

The system must provide explanation signals (e.g., skill- or keyword-level attribution) to support recruiter trust. Purely opaque black-box models without justification mechanisms will not be considered deployable solutions.

c. **Fairness and robustness checks.**

We will conduct controlled perturbation tests (e.g., varying gender-coded language while holding qualifications constant) to assess whether ranking behavior exhibits systematic bias. Fairness evaluation will be integrated into model assessment rather than treated as a post-hoc add-on.

10. These guardrails ensure that the evaluation framework reflects practical deployment realities and the broader business objective, rather than optimizing toward a narrow technical target.